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***Completed the project named as phase 2***

***NAME :Live Weather Dashboard***

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# Live Weather Dashboard – System Design

## Tech Stack Selection

- Frontend: React.js, TailwindCSS/Bootstrap,  
Axios/Fetch API, Recharts - Backend (Optional):  
Node.js + Express (for proxy & caching) - API  
Provider: OpenWeatherMap API /  
WeatherAPI.com - Deployment: Vercel / Netlify  
(Frontend), Render / Heroku (Backend)

## UI Structure

```
----- | Weather Dashboard
| ----- | Search Bar (Enter
city name) | [Search Button] | -----
----- | Current Weather Card: || - City Name, Country || -
Temperature (°C/°F) || - Weather Condition + Icon || -
Humidity, Wind Speed, Pressure | -----
----- | Forecast Section (Next 5 Days) || - Cards with
Date, Temp, Condition | -----
--- | Charts / Graphs || - Temperature Trend (line chart) || -
Humidity & Wind Speed (bar chart) | -----
-----
```

## API Schema Design

Endpoint: GET

`https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}` GET

`https://api.openweathermap.org/data/2.5/forecast?q={city}&appid={API_KEY}` Example Response (simplified): {  
"city": {"name": "Chennai", "country": "IN"}, "list": [ { "dt\_txt":  
"2025-09-15 12:00:00", "main": {"temp": 303.15, "humidity":  
78, "pressure": 1005}, "weather": [{"description": "light  
rain", "icon": "10d"}], "wind": {"speed": 4.5} } ] }

## Data Handling Approach

1. Frontend calls API with cityname input
2. API response parsed → Extract fields: City, Temp, Condition, Humidity, Wind, Pressure
3. State management with React useState/useEffect
4. Data mapped to UI Components (cards, charts)
5. Error handling with fallback UI
6. Optional caching in localStorage

## Component / Module Diagram

[App] | |-- [SearchBar  
Component] |--  
[CurrentWeatherCard] |--  
[ForecastList] | |--  
[ForecastCard] | |-- [Charts] |--  
[TemperatureChart] |--  
[HumidityWindChart]

## Basic Flow Diagram

UserInput (CityName)  
↓  
[SearchBar Component]  
↓  
API Call → Weather API  
↓  
JSON Response  
↓  
[Data Parser / State Update]  
↓  
UI Components Update  
↓  
User sees Weather Cards +  
Forecast + Charts